1 Greedy Block Selection

This report describes the greedy selection procedure in enough detail to reimplement it. The procedure takes a set of query blocks and a set of key blocks and returns a mask. It runs once per layer and its cost is dominated by a single pass over the block summaries. We deliberately keep the procedure free of learned parameters so that it can be dropped into an existing model.

1.1 Notation and Assumptions

We follow the notation of prior work and write block summaries with a bar over the symbol. Every block has the same number of positions, and the last block is padded when the sequence does not divide evenly. We assume that the summaries are computed before the selection pass and are available in fast memory. The mask is stored as a list of retained key block indices per query block rather than as a dense matrix.

Algorithm 1 Greedy block selection

```
M ← ∅
for each query block i do
    S ← TopK(q̄_i, K̄, k)
    M ← M ∪ {(i, j) : j ∈ S}
end for
return M
```

1.2 Cost of the Selection Pass

The selection pass compares every query summary with every key summary. This is quadratic in the number of blocks but the number of blocks is the sequence length divided by the block size. With a block size of sixty four the selection matrix is four thousand times smaller than the attention matrix. In wall clock terms the pass takes under two percent of the time of a dense attention call.

2 Correctness

The procedure never returns an empty mask because the diagonal block is always retained. It is deterministic given the summaries, so two runs on the same input produce the same mask. Ties in the top scoring set are broken by the block index, which keeps the output stable across hardware.

2.1 Padding and Masking

Padded positions receive a summary of zero and therefore never enter the retained set. Causal masking is applied after selection, so a retained block may contribute no positions at all. This is wasteful in principle but costs less than filtering the block list on the fly.

3 Practical Notes

The procedure is written for a single attention head and is launched once per head. Sharing the mask across heads saves time but costs about one point of accuracy in our tests. Sharing the mask across adjacent layers is almost free and we recommend it above eight thousand tokens. We did not find a setting in which sharing across both heads and layers was worth the loss.

4 Limitations

The report covers only the selection procedure and not the fused attention kernel. Numerical behaviour under half precision summaries is discussed in the companion paper. We have not evaluated the procedure on encoder decoder models.